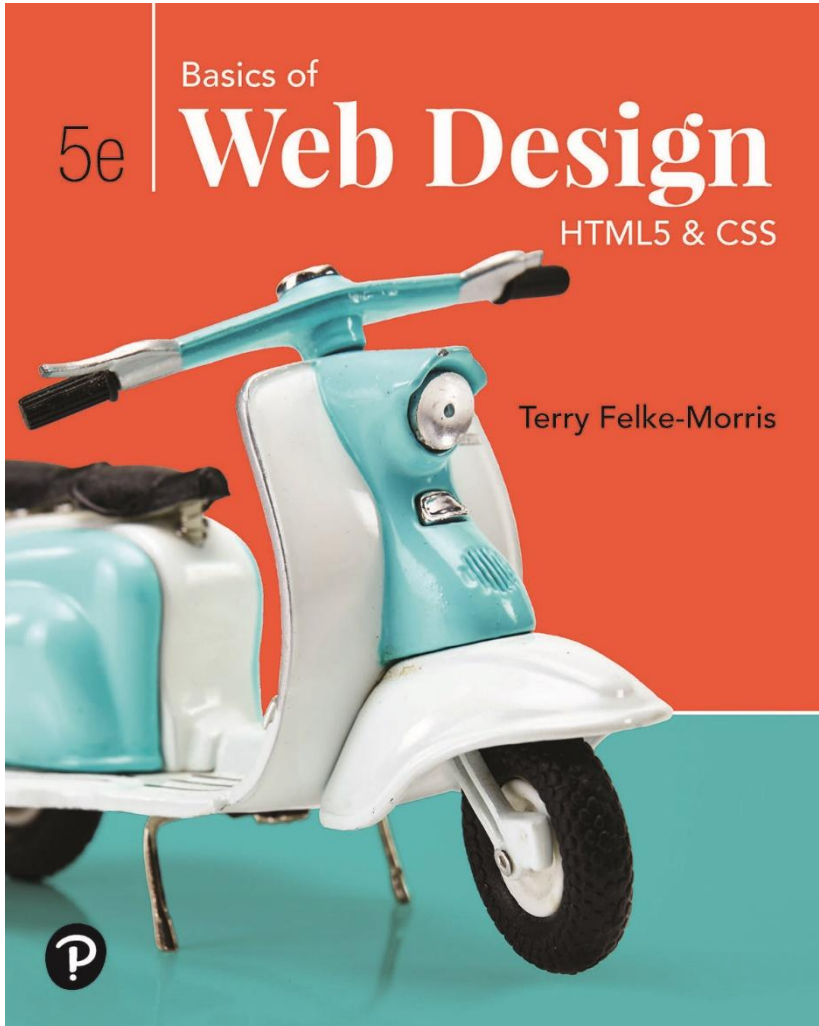


Basics of Web Design

Fifth Edition



Chapter 7

Page Layout Basics Key Concepts

Learning Outcomes

- Configure float
- Configure fixed positioning
- Configure relative positioning
- Configure absolute positioning
- Create two-column page layouts
- Configure vertical navigation in an unordered list
- Configure horizontal navigation in an unordered list.
- Add interactivity to hyperlinks with CSS pseudo-classes
- Configure CSS sprites
- Configure CSS for print
- Configure stacking order

Normal Flow

- Browser display of elements in the order they are coded in the Web page document

Figure 7.1

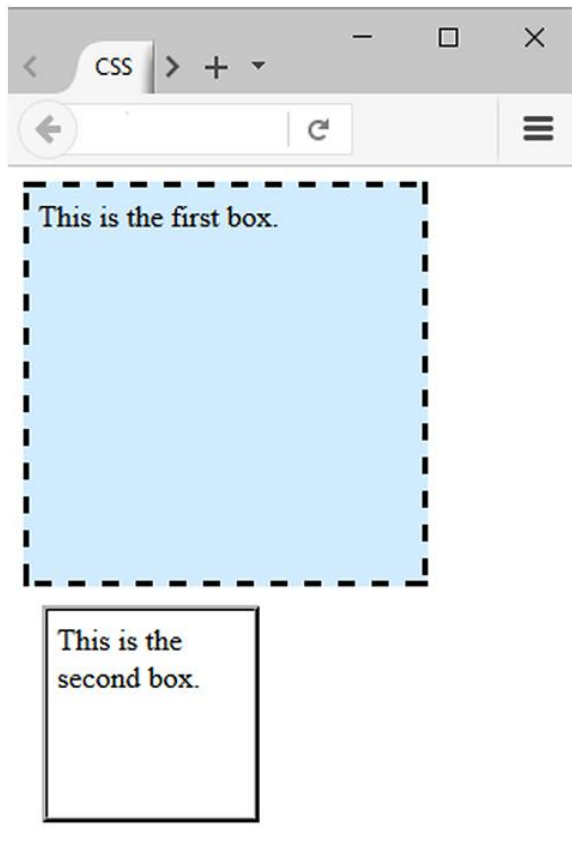
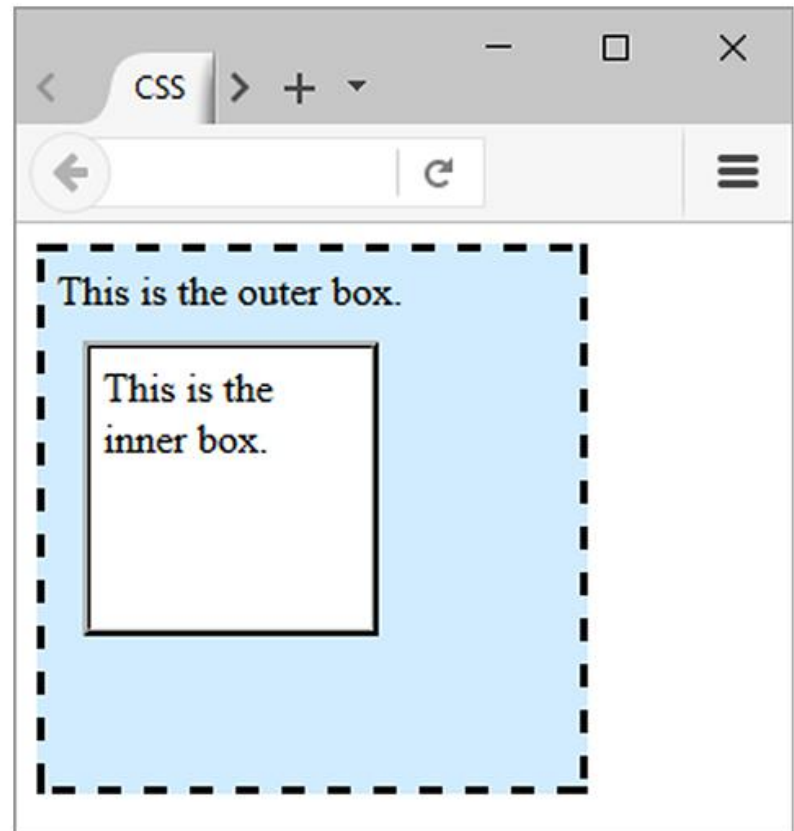
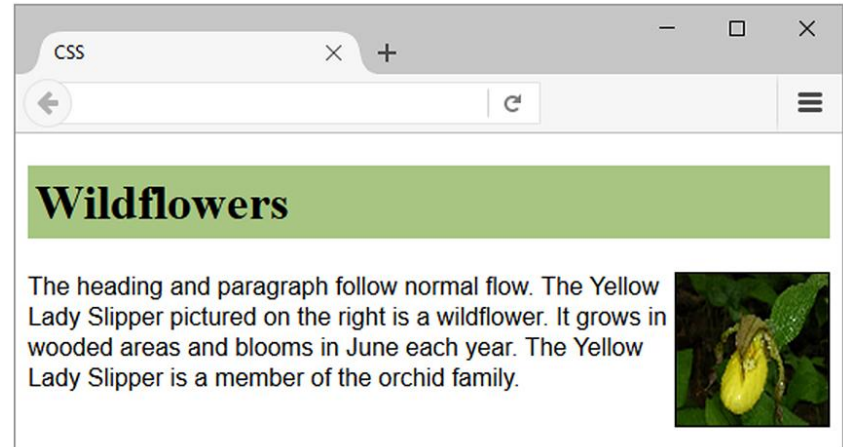


Figure 7.2



float Property

Elements that seem to “float” on the right or left side of either the browser window or another element are often configured using the CSS float property.



```
h1 { background-color: #A8C682;
```

```
padding: 5px;
```

```
color: #000000; }
```

```
p { font-family: Arial, sans-serif; }
```

```
#yls { float: right;
```

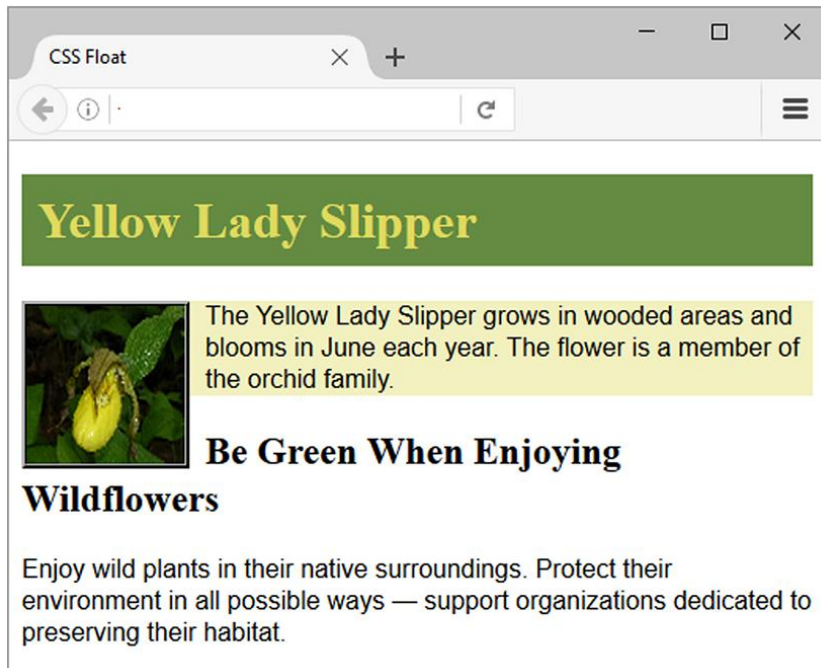
```
margin: 0 0 5px 5px;
```

```
border: solid; }
```

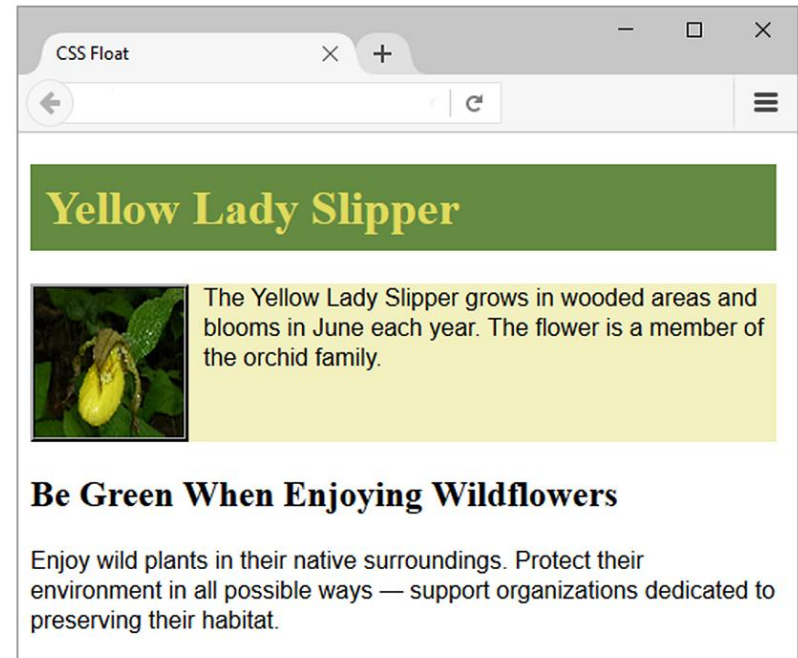
clear Property

- Useful to “clear” or terminate a float
- Values are left, right, and both

The h2 text is displayed in normal flow.



`clear: left;` was applied to the **h2**. Now the h2 text displays **AFTER** the floated image.

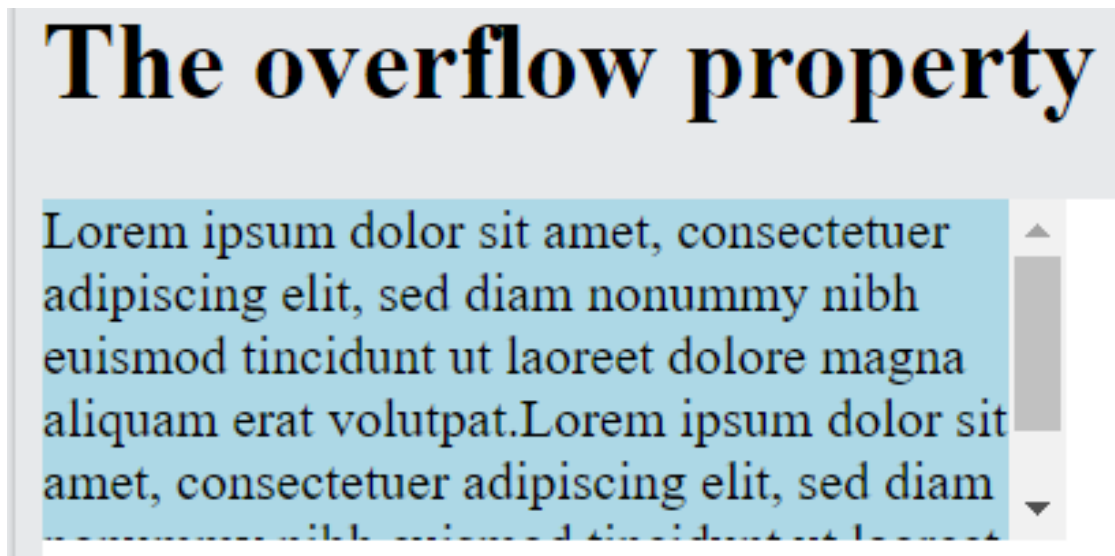


overflow Property

- Configures the display of elements on a web page.
- can occur when the content is too large to fit in the specified dimensions of the element. This property is essential for managing layout in web design
- Values are auto, hidden, and scroll

The background does not extend as far as you'd expect.

`overflow: auto;` was applied to the container **div**. Now the background extends and the h2 text displays AFTER the floated image.



The CSS box-sizing Property

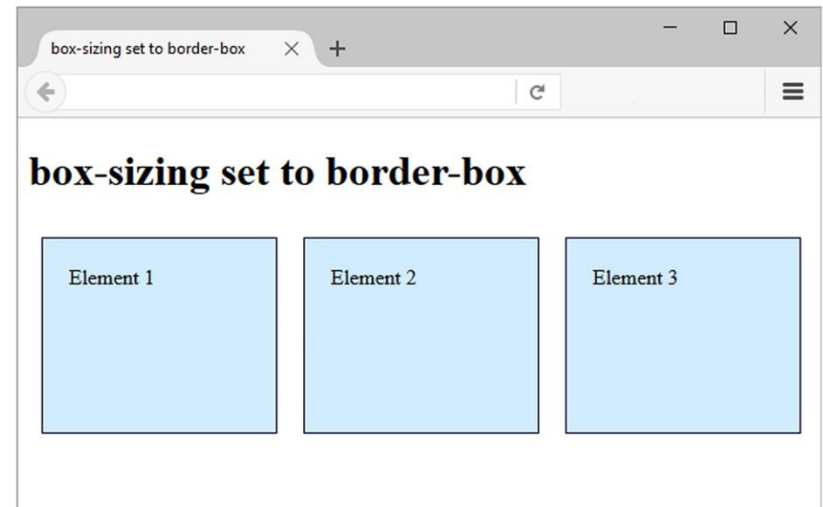
- Default value for width or height is the value for ONLY the content (not including border and padding).

The box-sizing property is used to selector to direct the browser to calculate the width and height of an element to include the value for content, padding, and border.

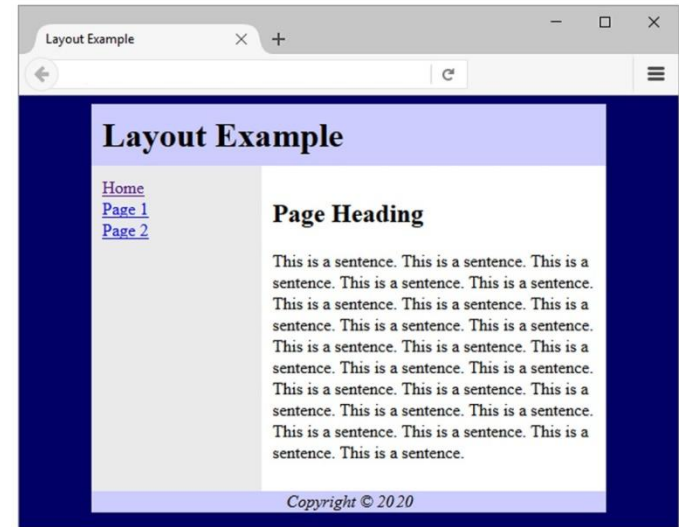
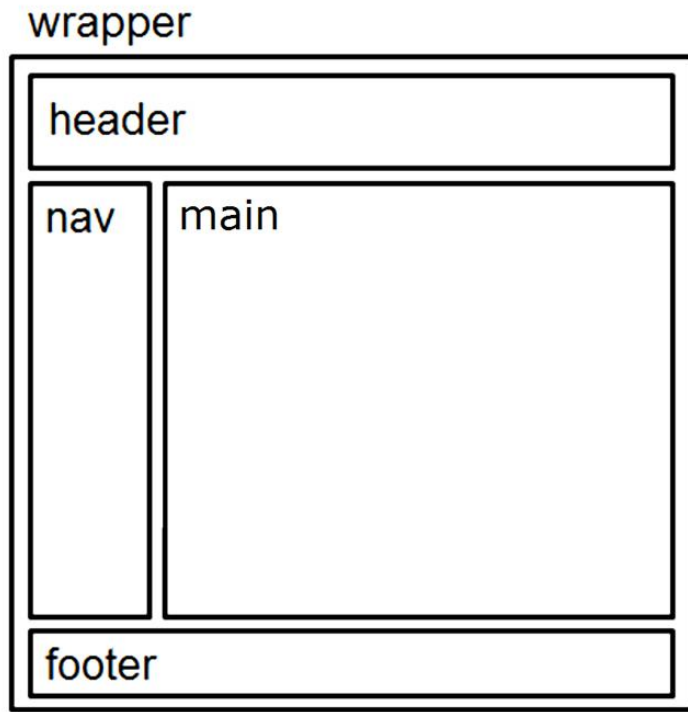
- Use the **universal selector** (*) to apply this to all the element on the page

Example:

```
* { box-sizing: border-box; }
```



Basic Two-Column Layout (1 of 2)



<body>

<div id="wrapper">

<header> <header>

<nav> </nav>

<main> </main>

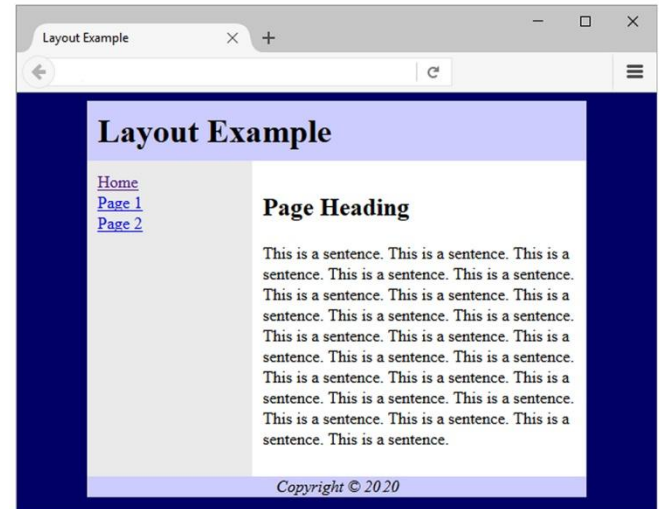
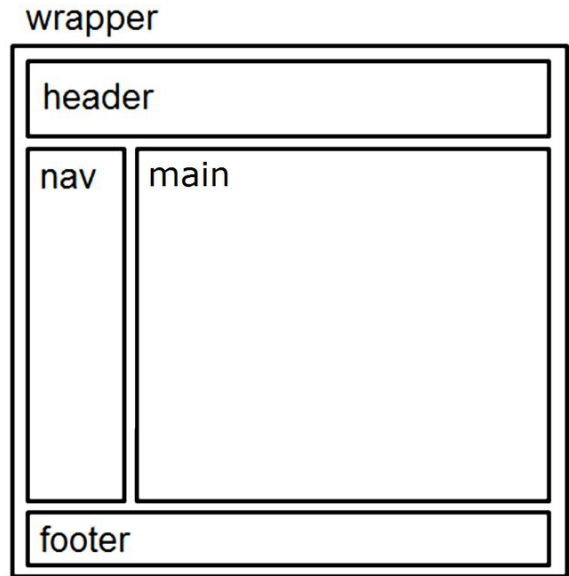
<footer> </footer>

</div>

</body>

Basic Two-Column Layout (2 of 2)

```
#wrapper { width: 80%;
            margin-left: auto;
            margin-right: auto;
            background-color: #EAEAEA; }
header { background-color: #CCCCFF; }
h1 { margin: 0; padding: 10px; }
nav { float: left;
      width: 150px;
      padding: 10px; }
main { margin-left: 160px;
      padding: 10px;
      background-color: #FFFFFF; }
footer { text-align: center;
        font-style: italic;
        background-color: #CCCCFF; }
```



Vertical Navigation with an Unordered List

```
<nav>
```

```
<ul>
```

```
<li><a href="index.html">Home</a></li>
```

```
<li><a href="menu.html">Menu</a></li>
```

```
<li><a
```

```
href="directions.html">Directions</a></li>
```

```
<li><a href="contact.html">Contact</a></li>
```

```
</ul>
```

```
</nav>
```

- CSS removes the list marker and underline:

```
nav ul { list-style-type: none; }
```

```
nav a { text-decoration: none; }
```

- 
- [Home](#)
 - [Menu](#)
 - [Directions](#)
 - [Contact](#)



[Home](#)
[Menu](#)
[Directions](#)
[Contact](#)

Display Property

- Configures how and if an element is displayed
 - **display: none;**
 - The element will not be displayed.
 - **display: block;**
 - The element is rendered as a block element even if it is actually an inline element, such as a hyperlink.
 - **display: inline;**
 - The element will be rendered as an inline element even if it is actually a block element, such as a .

Horizontal Navigation with an Unordered List

```
<nav>
```

```
<ul>
```

```
  <li><a href="index.html">Home</a></li>
```

```
  <li><a href="menu.html">Menu</a></li>
```

```
  <li><a href="directions.html">Directions</a></li>
```

```
  <li><a href="contact.html">Contact</a></li>
```

```
</ul>
```

```
</nav>
```

CSS removes the list marker, removes the underline, adds padding, and configures the list items for inline display.

```
nav ul { list-style-type: none;}
```

```
nav a { text-decoration: none;  
        padding-right: 10px; }
```

```
nav li { display: inline; }
```

A horizontal navigation bar with a light blue background. It contains four links: "Home", "Menu", "Directions", and "Contact", all in blue text.

Home Menu Directions Contact

CSS Pseudo-classes (1 of 2)

- Pseudo-classes and the anchor element
 - **:link** – default state for a hyperlink
 - **:visited** – a hyperlink that has been visited
 - **:focus** – triggered when the hyperlink has focus
 - **:hover** – triggered when the mouse moves over the hyperlink
 - **:active** – triggered when the hyperlink is being clicked

```
a:link {color:#000066;}
```

```
a:visited {color:#003366;}
```

```
a:focus {color:#FF0000;}
```

```
a:hover {color:#0099CC;}
```

```
a:active {color:#FF0000;}
```

CSS Pseudo-classes (2 of 2)

1. Text hyperlinks are underlined by default.



Print This Page

2. The `hover` pseudo-class is triggered by the mouse. The browser no longer displays the underline below the hyperlink.



Print This Page



```
a:link { color: #ff0000; }  
a:hover { text-decoration: none;  
          color: #000066; }
```

Position Property

Table 7.5 The `position` Property

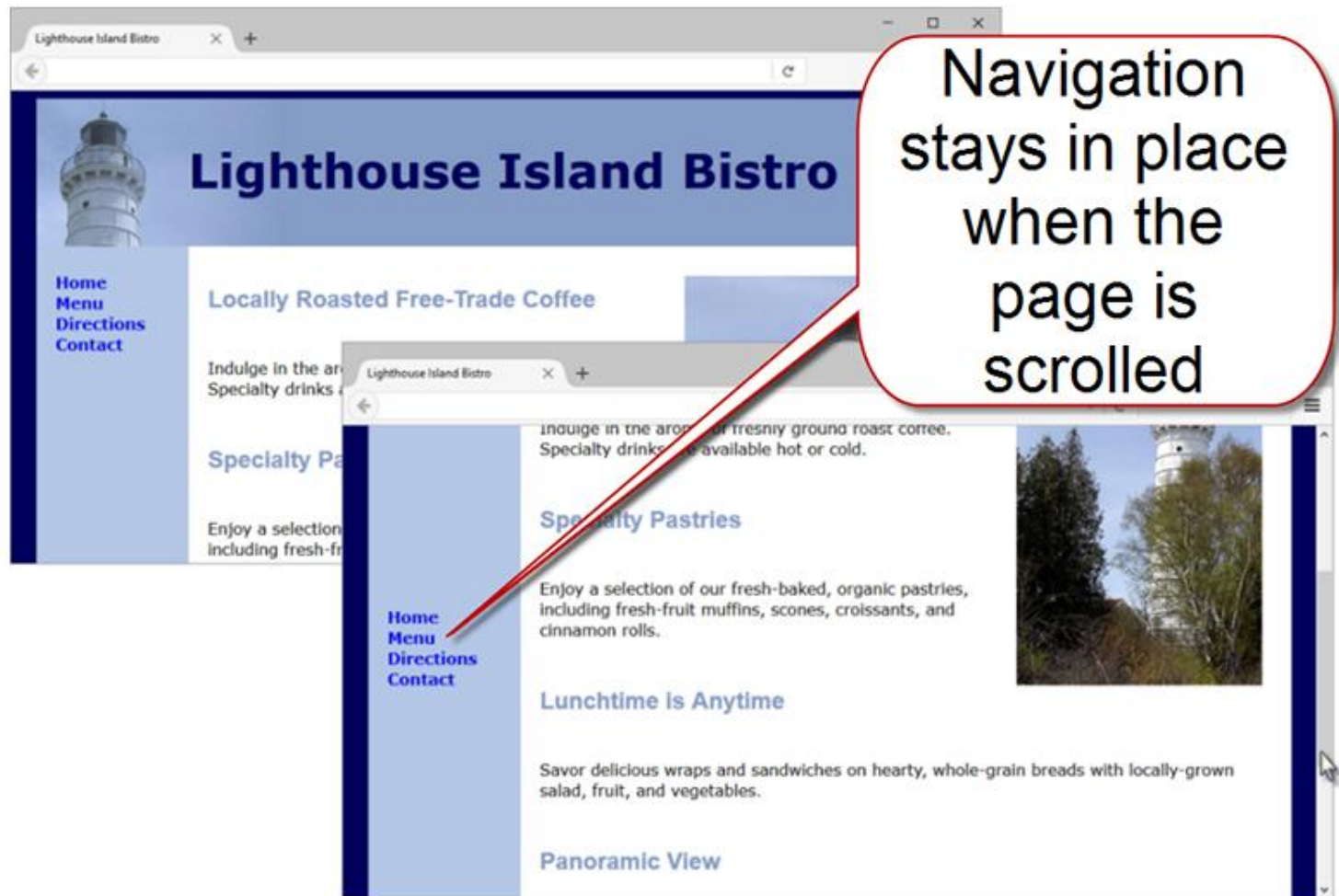
Value	Purpose
<code>static</code>	Default value; the element is rendered in normal flow
<code>fixed</code>	Configures the location of an element within the browser viewport; the element does not move when the page is scrolled
<code>relative</code>	Configures the location of an element relative to where it would otherwise render in normal flow
<code>absolute</code>	Precisely configures the location of an element outside of normal flow

Position Property

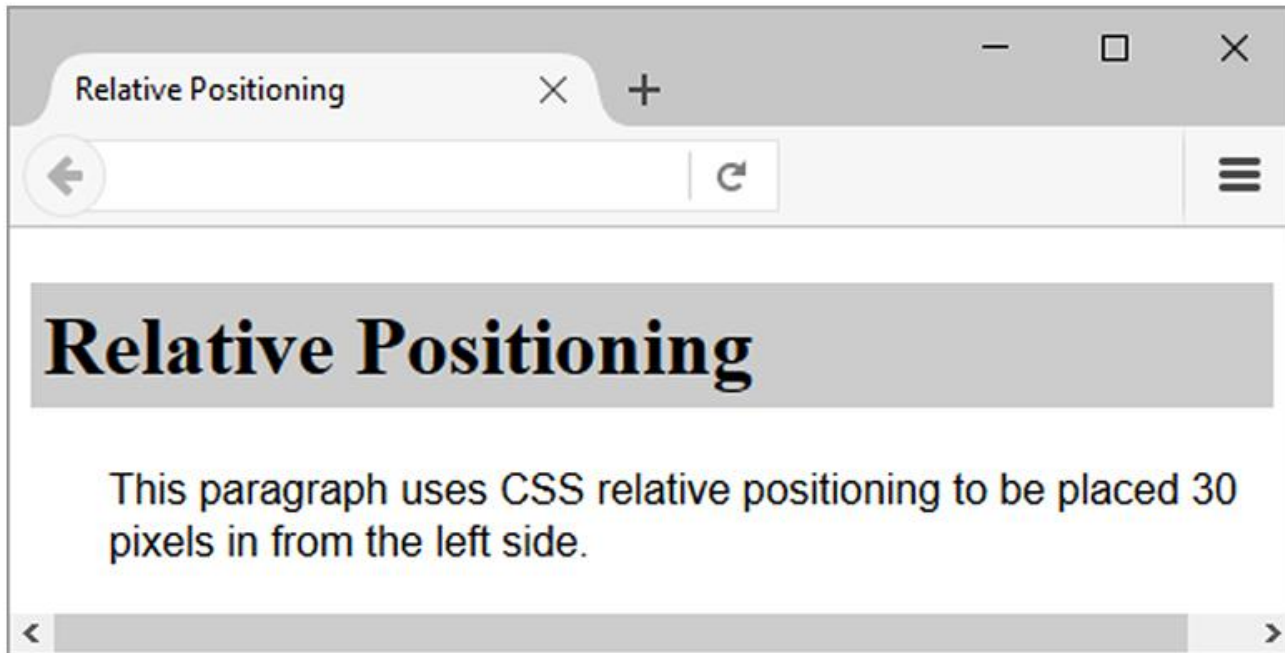
- Relative Positioning: Positions an element relative to its normal position in the document flow
- The element still takes up space in the normal flow, but can be shifted in any direction (up, down, left, right) from its original position using top, right, bottom, left properties
- Shifting the element does not affect the position of any other elements
- Absolute Positioning: Positions an element relative to its nearest positioned ancestor (parent element with a position value of relative, absolute, or fixed)
- The element is completely removed from the normal flow of the page
- **Relative positioning keeps the element in the normal flow, while absolute positioning removes it from the normal flow.**

Fixed Positioning

```
nav { position: fixed; }
```



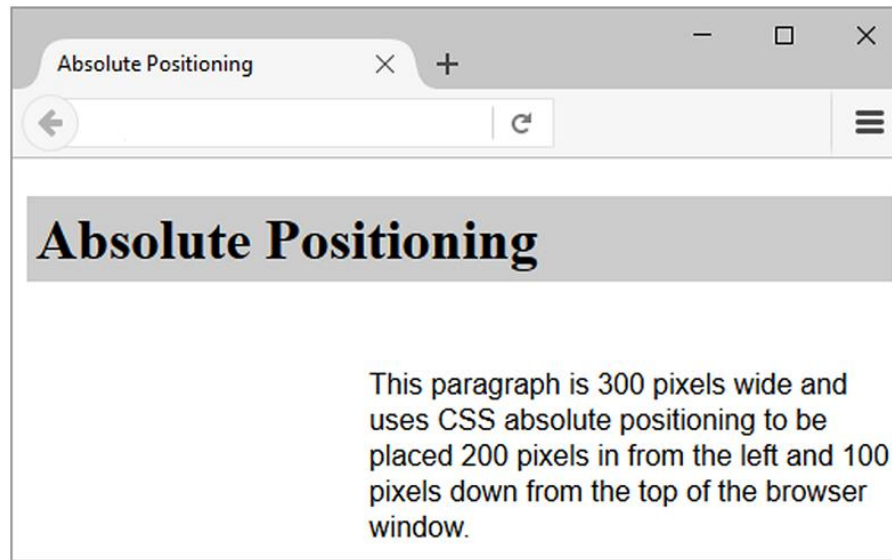
Relative Positioning



```
p { position: relative;  
    left: 30px;  
    font-family: Arial, sans-serif; }
```

Changes the location of an element in relation to where it would otherwise appear in normal flow

Absolute Positioning



```
p { position: absolute;  
    left: 200px;  
    top: 100px;  
    font-family: Arial, sans-serif;  
    width: 300px; }
```

Precisely specifies the location of an element outside of normal flow in relation to its first parent non-static element

CSS Debugging Tips

- Manually check syntax errors
- Use W3C CSS Validator to check syntax errors
 - <http://jigsaw.w3.org/css-validator/>
- Configure temporary background colors
- Configure temporary borders
- Use CSS comments to find the unexpected

/* the browser ignores this code */
- Don't expect your pages to look exactly the same in all browsers!
- Be patient!

Summary

- This chapter expanded your CSS skillset.
- You configured web pages with floating elements with CSS.
- You were introduced to fixed, relative, and absolute positioning.
- You configured web pages with two-column page layouts
- You used unordered lists to provide structure for navigation hyperlinks.
- You added interactivity to hyperlinks with CSS pseudo-classes.
- You configured a CSS sprite image.

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